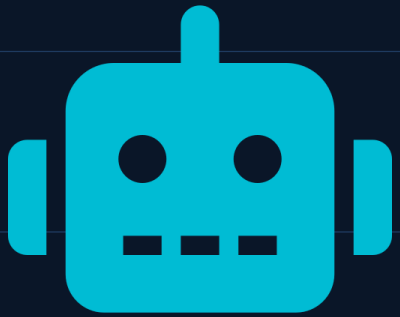
Finance Credit Follow-Up AI Agent



Automating Invoice Tracking · Escalation Management · Email Communication

Built With

Python · Streamlit · SQLite3

Domain

Finance · Accounts Receivable

Type

AI-Powered Automation Agent

Problem Statement

Manual Follow-Ups

Finance teams spend hours manually tracking overdue invoices and sending reminder emails one by one.

Inconsistent Communication

No standardized tone or escalation—early reminders sound the same as final notices, reducing effectiveness.

Poor Visibility & Tracking

Lack of centralized dashboards means overdue accounts slip through the cracks without timely action.

Compliance & Audit Risk

No structured audit trail of when, what, and who sent communications—creates regulatory and legal risk.

Businesses lose 10–15% of revenue annually due to delayed collections and ineffective follow-up processes.

Project Objective

AI-powered automation for finance collections

01

Automate Overdue Detection

Automatically calculate overdue days and classify invoices by risk level.

02

Intelligent Escalation

Assign escalation stages (1–5) based on overdue duration with appropriate communication tone.

03

AI Email Generation

Generate context-aware follow-up emails dynamically adapted to each escalation stage.

04

Complete Audit Trail

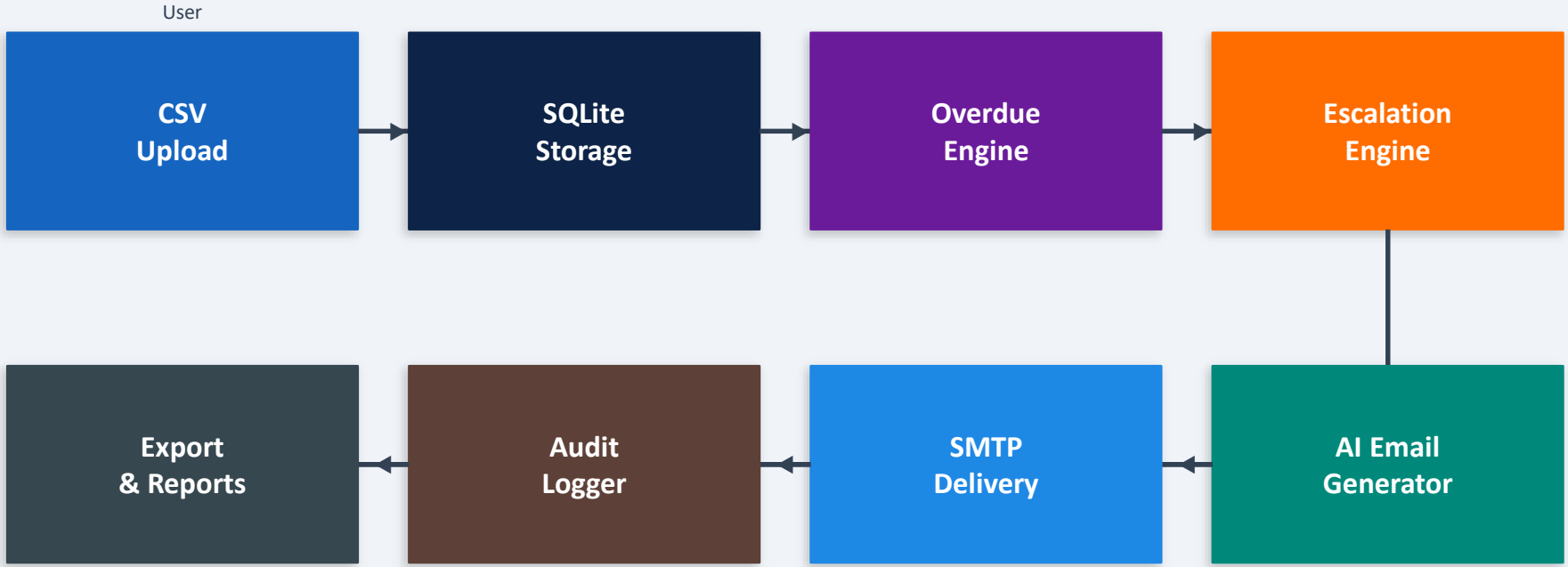
Log every action in SQLite—who, what, when—for compliance and review.

05

Real SMTP Delivery

Send actual emails via Gmail SMTP with dry-run simulation mode for safe testing.

System Architecture



Tech Stack

Technologies & Libraries Used

Frontend / UI

Streamlit

Web dashboard & interactive interface

Backend

Python 3.x

Core logic, data processing, orchestration

Database

SQLite3

Persistent storage — invoices & audit logs

Data Processing

Pandas · Datetime

CSV parsing, overdue calculation, date ops

Email Sending

smtplib · email.mime

Gmail SMTP with App Password authentication

Visualization

Plotly

Interactive charts and dashboard metrics

AI / Logic Engine

Template-Based AI

Escalation-aware dynamic email generation

Export Features

CSV · JSON Export

Downloadable audit and invoice reports

Escalation Workflow

Stage 1	Stage 2	Stage 3	Stage 4	Escalate
1–7 Days	8–14 Days	15–21 Days	22–30 Days	30+ Days
Friendly Reminder	**Polite but Firm**	**Formal & Serious**	**Final Reminder**	**Legal Review**
Polite, helpful tone. Payment may have been overlooked.	Acknowledges delay, requests immediate action.	Professional urgency, references contract terms.	Last warning before escalation to management.	Flagged for manual finance/legal team intervention.
Risk: Low	Risk: Medium	Risk: High	Risk: Critical	Risk: Escalate

Dashboard Features

Total Invoices

Overdue Amount

Emails Sent

High Risk Accounts

CSV Upload & Parse

Upload invoices via drag-and-drop interface with auto-validation

Smart Filters

Filter by stage, risk level, date range, client name

AI Email Generator

One-click generation of tone-appropriate follow-up emails

Manual Overrides

Human-in-the-loop edits before sending — full control retained

Plotly Charts

Interactive bar, pie, and trend charts for overdue analysis

CSV / JSON Export

Download full invoice or audit log reports in your preferred format

Database & Audit Logging

invoices

```
1  invoice_no
2  client_name
3  amount
4  due_date
5  email
6  overdue_days
7  stage
8  risk_level
9  send_status
```

audit_logs

```
1  id (PK)
2  timestamp
3  invoice_no
4  action
5  old_stage
6  new_stage
7  reason
```

Full compliance audit trail

Stage change history tracked

Persistent across sessions

Exportable for review

Real Email Sending via SMTP

1. **AI Generates Email**
   Template engine creates tone-aware email body based on escalation stage
2. **User Reviews & Approves**
   Human-in-the-loop review with manual override capability before sending
3. **Mode Selection**
   Choose between Dry Run (simulation) or Live Send (real SMTP delivery)
4. **SMTP Authentication**
   Connects to Gmail SMTP (smtp.gmail.com:587) using App Password (TLS)
5. **Email Delivered**
   email.mime constructs headers; smtplib sends; send_status updated in DB

Sending Modes

Dry Run Mode

- • Simulates email sending
- • No actual email delivered
- • Safe for testing & demos

Live Send Mode

- • Real Gmail SMTP delivery
- • Requires App Password config
- • Status tracked in database
- • Audit log entry created

Conclusion & Future Scope

Project Achievements

- ✓ End-to-end invoice collection automation
- ✓ 5-stage intelligent escalation engine
- ✓ AI-generated tone-aware follow-up emails
- ✓ Gmail SMTP real email delivery
- ✓ Full SQLite persistence & audit logging
- ✓ Interactive Streamlit dashboard with Plotly charts
- ✓ Human-in-the-loop design for manual overrides
- ✓ CSV / JSON export for reporting

🚀 Future Scope

Gemini / OpenAI Integration
Replace template logic with true LLM-powered email generation

ML Risk Prediction
ML model to predict payment probability per client

Automated Scheduling
APScheduler for daily auto-dispatch of follow-ups

Payment Portal Link
Embedded payment link in follow-up emails

Multi-User Auth
Role-based access for finance team members